

Blocking Caspase-3-Mediated Proteolysis of IKK β Suppresses TNF- α -Induced Apoptosis

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Summary

The transcription factor NF- κ B is essential for survival of many cell types. However, cells can undergo apoptosis despite the concurrent NF- κ B activation. It is unknown how the protection conveyed by NF- κ B is overridden during apoptosis. We report here that I κ B kinase (IKK) β was specifically proteolyzed by Caspase-3-related caspases at aspartic acid residues 78, 242, 373, and 546 during tumor necrosis factor (TNF)- α -induced apoptosis. Proteolysis of IKK β eliminated its enzymatic activity, interfered with IKK activation, and promoted TNF- α killing. Point mutations that abrogate IKK β proteolysis generated a caspase-resistant IKK β mutant, which suppressed TNF- α -induced apoptosis. Thus, our study demonstrates that TNF- α -induced apoptosis requires caspase-mediated proteolysis of IKK β .

Introduction

The I κ B kinase complex (IKK) is a multiple protein complex with a molecular weight of 700–900 kDa (Lee et al., 1997). The IKK complex has two catalytic subunits, IKK α and IKK β (IKK-1 and IKK-2) (DiDonato et al., 1997; Regnier et al., 1997; Zandi et al., 1997; Woronicz et al., 1997; Mercurio et al., 1997), and an essential regulatory subunit NEMO/IKK γ /IKKAP1 (Yamaoka et al., 1998; Rothwarf et al., 1998; Mercurio et al., 1999). IKK can be activated by a variety of stimuli, from proinflammatory cytokines to physical stress, through protein phosphorylation (Mercurio et al., 1997; Nemoto et al., 1998; Karin and Ben-Neriah, 2000). Once activated, IKK phosphorylates I κ Bs, a group of cytoplasmic inhibitors of NF- κ B, on specific serines (Ser-32 and -36 in I κ B α and Ser-19 and -21 in I κ B β), triggering their ubiquitination and subsequent degradation by the 26S proteasome (Karin and Ben-Neriah, 2000). Degradation of I κ B unmasks NF- κ B's nuclear translocation signals. This allows NF- κ B to translocate into the nucleus where it stimulates transcription of specific target genes that are involved in immune responses, inflammation, viral infection, and cell survival (Siebenlist et al., 1994; Thanos and Maniatis, 1995; Verma et al., 1995; Baeuerle and Baltimore, 1996; Baldwin, 1996; Barnes and Karin, 1997; Ghosh et al., 1998).

Compelling genetic evidence shows that in many cell

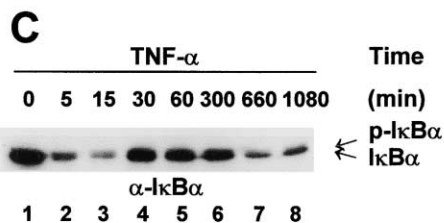
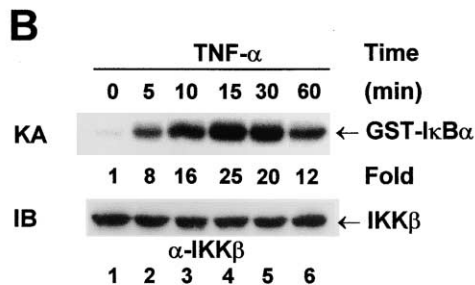
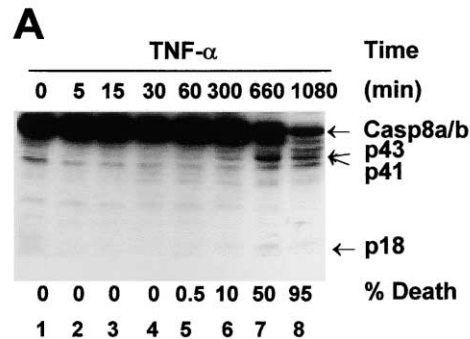
types the IKK/NF- κ B pathway is required for cell survival. Genetic disruption experiments in mice reveal that RelA/NF- κ B-deficient mice die around embryonic day 15, resulting from massive apoptosis of hepatocytes in the liver (Beg and Baltimore, 1996). Consistently, disruption of IKK β (Li et al., 1999b; Tanaka et al., 1999; Li et al., 1999c) or IKK γ (Makris et al., 2000; Schmidt-Supprian et al., 2000; Rudolph et al., 2000) alleles in mice severely impairs NF- κ B activation by proinflammatory stimuli, such as TNF- α , resulting in embryonic lethality with massive apoptosis in the liver. In contrast, disruption of IKK α alleles only slightly affects NF- κ B activation, although the mice die of perinatal lethality with severe morphogenetic defects in keratinocyte proliferation and differentiation (Hu et al., 1999; Takeda et al., 1999; Li et al., 1999a). Biochemical data also show that NF- κ B controls expression of several antiapoptotic proteins, including inhibitors of apoptosis (c-IAP1, c-IAP2, and XIAP) and TNF receptor-associated factors (TRAF1 and TRAF2) (Chu et al., 1997; Wang et al., 1998; Wu et al., 1988). Indeed, overexpression of NF- κ B suppresses apoptosis (Liu et al., 1996; Beg and Baltimore, 1996; Van Antwerp et al., 1996). Furthermore, the I κ B α (32/36 AA) mutant, which can no longer be phosphorylated and is therefore resistant to ubiquitin-mediated degradation, suppresses NF- κ B activation and sensitizes cells to apoptotic insults (Wang et al., 1998).

The proinflammatory cytokine TNF- α exerts its biological activity by binding to type 1 and type 2 receptors (TNF-R1 and TNF-R2) and activating multiple signaling pathways in many cell types (Tartaglia and Goeddel, 1992; Tracey and Cerami, 1993). The TNF-R1 signaling complex is composed of the trimerized receptor, TNF-R1-associated death domain protein (TRADD), FAS-associated death domain protein (FADD), TRAF2, and receptor interacting protein (RIP) (Locksley et al., 2001). FADD recruits and activates pro-Casp8 (Muzio et al., 1996), initiating the apoptotic pathway in which Casp3 and Casp7 are two major effector caspases (Cryns and Yuan, 1998). Activated Casp8 also cleaves Bid (BH3-interaction domain death agonist) (Wang et al., 1996; Li et al., 1998; Luo et al., 1998), which releases cytochrome C from mitochondria to induce apoptosis (Zha et al., 2000; Budihardjo et al., 1999; Green and Reed, 1998). In contrast, TRAF2 and RIP are involved in activation of c-Jun N-terminal kinase (JNK) and IKK (Liu et al., 1996; Ashkenazi and Dixit, 1998), resulting in activation of c-Jun and NF- κ B, respectively. Although the effect of JNK/c-Jun activation on TNF- α -induced apoptosis requires further investigation, activation of IKK and NF- κ B suppresses TNF- α -induced apoptosis in most cell types (Ashkenazi and Dixit, 1998). However, it is poorly understood how the protection conveyed by the IKK/NF- κ B pathway is overridden during TNF- α -induced apoptosis. We report here that IKK β is proteolyzed by Casp3-related caspases during TNF- α -induced apoptosis, and a caspase-resistant IKK β mutant suppresses TNF- α killing.

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MCF-7 cells



HeLa cells

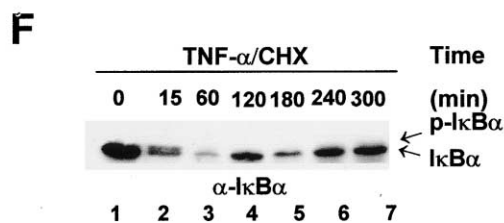
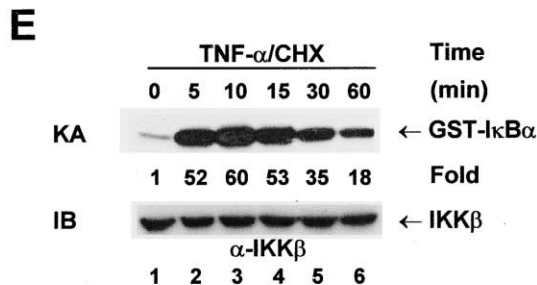
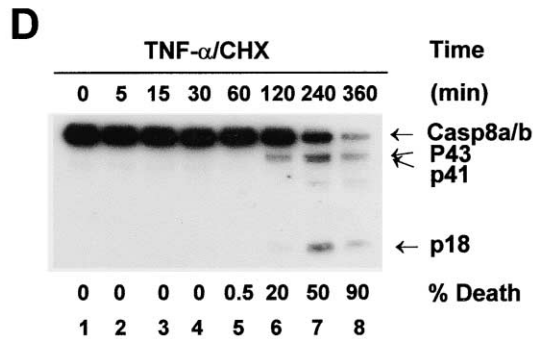


Figure 1. Cells Undergo Apoptosis despite the Concurrent Activation of IKK and NF- κ B

MCF-7 or HeLa cells were treated with TNF- α (10 ng/ml) or TNF- α (10 ng/ml) and CHX (10 μ g/ml) for times indicated.

(A and D) Proteolysis of Casp8 was analyzed by immunoblotting using anti-Casp8 antibody. Cell viability was monitored by trypan blue exclusion.

(B and E) IKK was isolated by immunoprecipitation with anti-IKK β antibody and its activity was measured by immunocomplex kinase assay (KA) using purified GST-IkB α (1–54) (2 μ g) as substrate. An aliquot of each lysate was analyzed for its content of IKK β by immunoblotting (IB) using anti-IKK β C-terminal antibody.

(C and F) Degradation and resynthesis of I kB α proteins were analyzed by immunoblotting using anti-I kB α antibody. These experiments were repeated three times with similar results.

Results

IKK β Is Specifically Cleaved during TNF- α -Induced Apoptosis In Vivo

TNF- α can stimulate both pro- and antiapoptotic signaling pathways that determine life and death of a cell. TNF- α -induced apoptosis in a subline of human breast carcinoma MCF-7 cells, as monitored by activation of Casp8 and cell viability assay (Figure 1A). In addition, TNF- α also stimulated IKK (Figure 1B), resulting in degradation of I kB α (Figure 1C). Considering that I kB α is a NF- κ B target gene (Le Bail et al., 1993), the rapid reappearance of the newly synthesized I kB α proteins indicated that NF- κ B was activated (Figure 1C). TNF- α also induced apoptosis in human cervical carcinoma HeLa cells in the presence of the protein synthesis inhibitor cycloheximide (CHX) (Figure 1D), which inhibits the synthesis of antiapoptotic proteins induced by NF- κ B

(Miura et al., 1995; Wang. et al., 1998). At the concentration of 10 μ g/ml used here, CHX reduced, but did not completely block protein synthesis, as evident by the slower reappearance of newly synthesized I kB α proteins (Figure 1F). Despite the induction of apoptosis, IKK and NF- κ B were still rapidly activated after TNF- α stimulation (Figures 1E and 1F). These results show that cells were able to undergo apoptosis in spite of the concurrent activation of IKK and NF- κ B.

Activation of IKK and NF- κ B suppresses TNF- α killing (Karin and Ben-Neriah, 2000). Cells have to override the protection conveyed by the IKK/NF- κ B pathway to carry out apoptosis. It is possible that cells may destroy IKK during apoptosis. Indeed, immunoblotting analysis using an antibody against the unique carboxyl (C) region of IKK β (amino acid 629–735) revealed that IKK β was completely degraded in MCF-7 cells (Figure 2A) and significantly in HeLa cells (Figure 2B) during TNF- α -

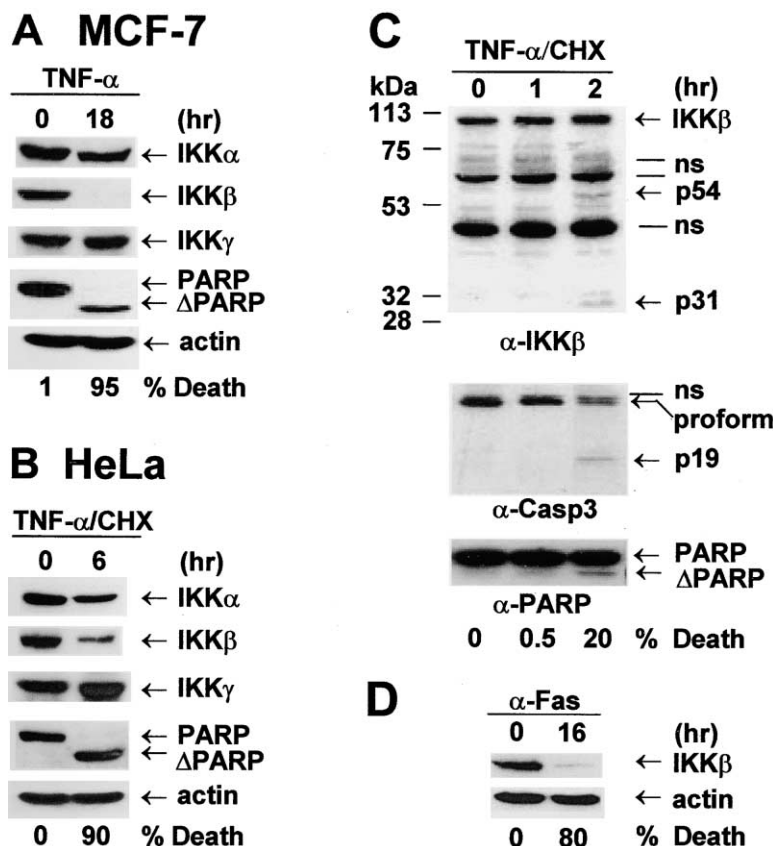


Figure 2. IKK β but Not IKK α and IKK γ Was Proteolyzed during Apoptosis Induced by TNF- α or Anti-Fas Antibody

MCF-7 or HeLa cells were treated as described in Figure 1.

(A and B) Immunoblotting was performed with specific antibodies against IKK α , the C terminus of IKK β , IKK γ , PARP, and actin, as indicated. Cell viability was analyzed by trypan blue exclusion.

(C) HeLa cells were treated with TNF- α (10 ng/ml) and CHX (10 μ g/ml) for times indicated. Immunoblotting was performed with antibodies against (top) the N terminus of IKK β , (middle) Casp3, and (bottom) PARP. p54 and p31 were two *in vivo*-proteolytic fragments of IKK β . Cell viability was analyzed by trypan blue exclusion. ns, nonspecific. p19 is the active form of Casp3.

(D) HeLa cells were treated with anti-Fas antibody for 16 hr, and immunoblotting was performed with the anti-IKK β C-terminal antibody. These experiments were repeated twice with similar results.

induced apoptosis. In contrast, the proteolysis was marginal for IKK α and undetectable for IKK γ (Figures 2A and 2B). Cell death was monitored by degradation of PARP, which is an authentic substrate of Casp3, and by cell viability assay (Figures 2A and 2B).

Immunoblotting analysis using an antibody against the first 12 amino acids of IKK β 's amino (N) terminus revealed two major proteolytic fragments with molecular weights of 54 (p54) and 31 kDa (p31) in TNF- α /CHX-treated HeLa cells (Figure 2C). Preincubation of the IKK β N-terminal antibody with its peptide antigen abolished its ability to recognize both p54 and p31, as well as intact IKK β (data not shown), suggesting that p54 and p31 were indeed fragments of IKK β . The p54 and p31 fragments appeared within 2 hr of TNF- α /CHX treatment, corresponding to the initial activation of Casp3 and cleavage of PARP (Figure 2C, lane 3). In MCF-7 cells, the kinetics of IKK β proteolysis corresponded to activation of Casp7 (MCF-7 cells have no detectable Casp3) (Janicke et al., 1998) and PARP cleavage (data not shown), although the major proteolytic fragment was p31 (data not shown).

Fas is another member of the TNF receptor superfamily, and occupation of Fas by Fas ligand (FasL) or anti-Fas antibody induces apoptosis in many cell types (Tartaglia and Goeddel, 1992; Locksley et al., 2001). Since inhibition of NF- κ B promotes apoptosis induced by Fas-ligation (Dudley et al., 1999), cells may need to eliminate the IKK/NF- κ B pathway for Fas-mediated apoptosis. Indeed, immunoblotting analysis using the anti-IKK β C-terminal antibody revealed that IKK β was proteolyzed

in HeLa cells treated with anti-Fas antibody (Figure 2D), as in TNF- α -induced apoptosis (Figures 2A and 2B).

IKK β Is Cleaved by Casp3-Related Caspases *In Vivo* and *In Vitro*

The proteolysis of IKK β during TNF- α -induced apoptosis prompted us to investigate whether IKK β was cleaved by caspases, a group of cysteine proteases that are critical for apoptotic cell death (Cohen, 1997; Cryns and Yuan, 1998; Thornberry and Lazebnik, 1998). HeLa cells were treated with z-VAD-FMK (a general caspase inhibitor), DEVD-FMK (a caspase inhibitor highly specific for Casp3 and related caspases like Casp7), Ac-YVAD-CMK (a caspase inhibitor highly specific for Casp1) (Cohen, 1997), or mock-treated for 1 hr, followed by treatment with TNF- α /CHX for 6 hr. Immunoblotting analysis with the anti-IKK β C-terminal antibody revealed that proteolysis of IKK β (Figure 3A) was completely blocked by z-VAD or DEVD (Figure 3A), but not YVAD (Figure 3A), mirroring the degradation of PARP (Figure 3A).

The proteolysis of IKK β was further examined using an *in vitro* proteolysis assay. *In vitro* translated 35 S-labeled HA-IKK β was cleaved by apoptotic, but not control extracts, producing four major fragments with molecular weights of 98, 78, 54, and 31 kDa (Figure 3B). Among them, both 78 and 31 kDa fragments were doublets (also see Figure 4B). The 54 and 31 kDa fragments matched the *in vivo* fragments p54 and p31 (Figure 2C), further supporting the notion that p54 and p31 were *in vivo* proteolytic fragments of IKK β . Additionally, HA-IKK β was found to be converted to IKK β after the cleavage

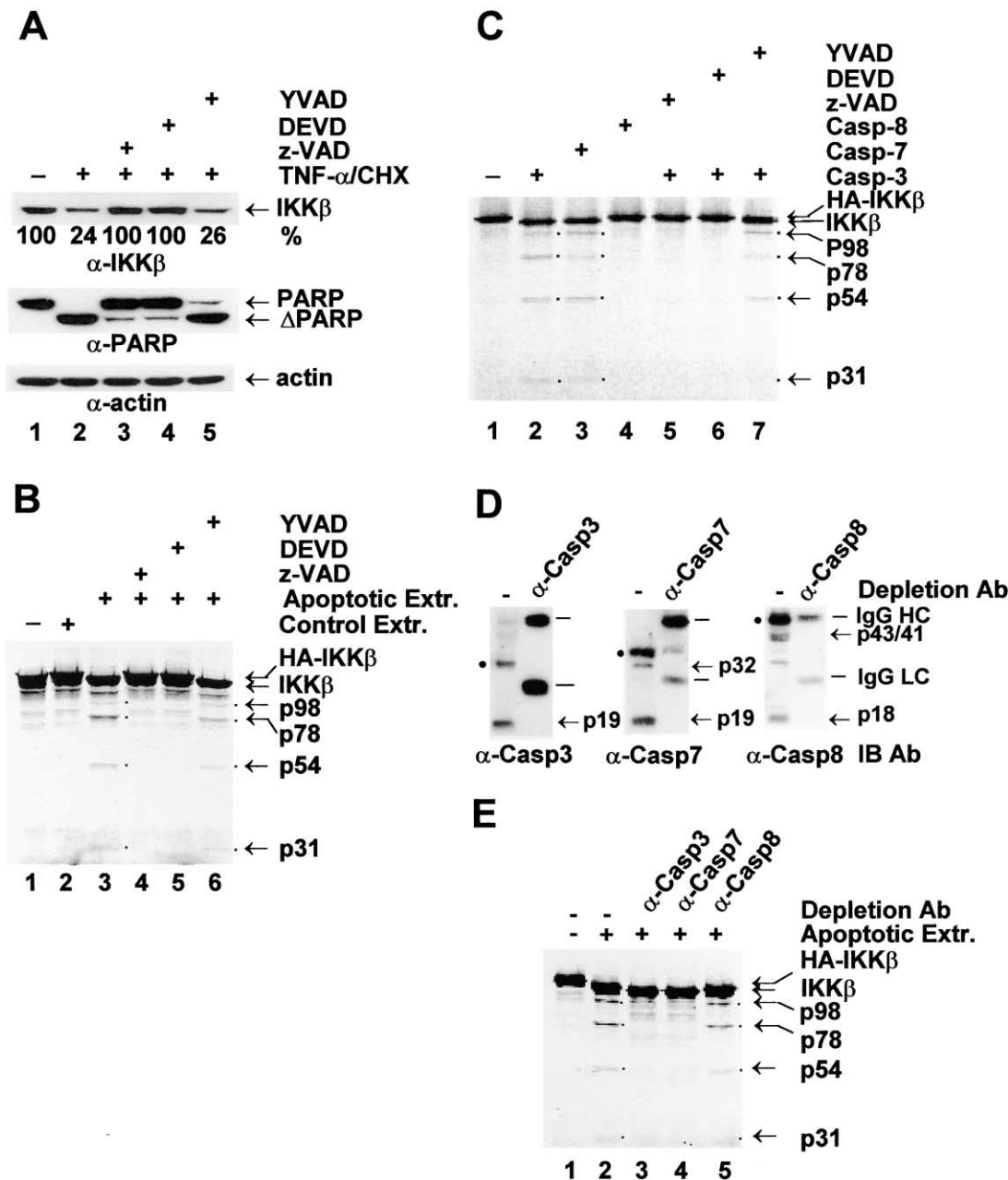


Figure 3. Proteolysis of IKK β by DEVD-Sensitive Casp3 and Casp7 but Not Casp8 In Vivo and In Vitro

(A) HeLa cells were pretreated with caspase inhibitors z-VAD-FMK (10 μ M), DEVD-FMK (1 μ M), or Ac-YVAD-CMK (1 μ M) or mock treated for 1 hr. After treatment with TNF- α (10 ng/ml) and CHX (10 μ g/ml) for 6 hr, cells were harvested, and cell extracts were analyzed by immunoblotting using antibodies against the C terminus of IKK β , actin, or PARP.

(B) 35 S-labeled HA-IKK β proteins were incubated with control, apoptotic HeLa extracts, or apoptotic extracts pre incubated with various caspase inhibitors as described in (A) or mock treated.

(C) 35 S-labeled HA-IKK β proteins were incubated with purified Casp3, Casp7, Casp8, or Casp3 preincubated with various caspase inhibitors as described in (A).

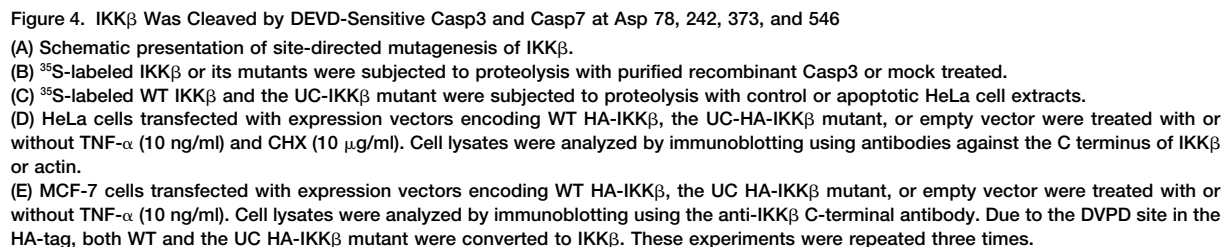
(D) Casp3, Casp7, and Casp8 were immunodepleted from apoptotic HeLa cell extracts. The proforms of caspases are indicated by dots, while their active forms are indicated by arrows. IgG heavy chain (HC) and light chain (LC) are indicated by bars.

(E) 35 S-labeled HA-IKK β proteins were incubated with mock-treated or caspase-depleted apoptotic HeLa cell extracts. These experiments were repeated three times with similar results.

of its HA tag, which contains a DVPD caspase site (Figure 3B). The proteolysis of HA-IKK β was blocked when the apoptotic extracts were preincubated with z-VAD or DEVD, but not YVAD (Figure 3B).

To test whether IKK β was directly cleaved by DEVD-

sensitive caspases such as Casp3 and Casp7, 35 S-labeled HA-IKK β was subjected to proteolysis in vitro with purified recombinant Casp3 or Casp7. DEVD-insensitive Casp8 was used as a control. Casp3 or Casp7 cleaved HA-IKK β into four major proteolytic fragments



To determine whether Casp3 and/or Casp7 were the caspases that cleaved IKK β in vivo, Casp3, Casp7, or Casp8 were immunodepleted from apoptotic HeLa cell extracts with their corresponding antibody (Figure 3D). Depletion of Casp3 or Casp7 significantly blocked the cleavage of ³⁵S-labeled HA-IKK β by apoptotic HeLa ex-

Caspases are known to cleave their substrates at the carboxyl terminus of specific aspartic acids (Asp, D) (Cryns and Yuan, 1998; Thornberry and Lazebnik, 1998). The Asp and three additional amino acids that precede it form a relatively conserved caspase cleavage motif

but exceptions do exist. The conserved cleavage motif for Casp3 and Casp7 is DXVD (Cohen, 1997), where X is any amino acid. Although IKK β was cleaved by Casp3 or Casp7 into multiple fragments (Figure 3B), only one such Casp3/7 cleavage motif, DIVD⁵⁴⁶, was found in IKK β . Thus, mutation sites were designed based on the size of proteolytic fragments of IKK β . The Asp residues at potential cleavage sites were mutated to noncleavable amino acids (alanine, Ala, A; or glutamic acid, Glu, E) by site-directed mutagenesis (Figure 4A). The resultant mutants were *in vitro* translated in the presence of ³⁵S-methionine and subjected to proteolysis using Casp3 (Figure 4B).

The major proteolytic fragments of IKK β were p98, a p78/p76 doublet, p54, and a p31/p30 doublet (Figure 4B, lane 2). Replacement of Asp at position 546, but not 493 or 508 (data not shown), with Ala eliminated p78(1–546) and p30(547–756) (Figure 4B, lane 4). Replacement of Asp373, but not Asp383 (data not shown), with Ala in the IKK β (D546A) mutant eliminated p54 (Figure 4B, lane 6), suggesting that p54 is likely to be a doublet of fragments 1–373 and 374–756 since they have almost identical molecular weights. Further replacement of Asp 242, but not Asp204 or Asp730 (data not shown), with Ala in the IKK β (D373A/D546A) mutant eliminated p76(243–756) and p31(1–242) (Figure 4B, lane 8). Finally, replacement of Asp78, but not Asp90 (data not shown), with Glu in the IKK β (D242A/D373A/D546A) mutant eliminated p98(79–756) (Figure 4B). The resultant IKK β (D78E/D242A/D373A/D546A) mutant was also resistant to Casp7 cleavage *in vitro* (data not shown). Thus, it was termed the uncleavable IKK β (UC IKK β) mutant. Proteolysis assays revealed that while wild-type (WT) HA-IKK β was proteolyzed by apoptotic HeLa extracts (Figure 4C, lane 3), the UC IKK β mutant was resistant (Figure 4C, lane 6).

To determine whether the UC IKK β mutant is resistant to proteolysis during apoptosis, HeLa cells were transiently transfected with expression vectors encoding WT HA-IKK β , the UC HA-IKK β mutant, or empty vector. Cells were treated with TNF- α /CHX or were left untreated. Immunoblotting analysis with the anti-IKK β C-terminal antibody revealed that expression of WT IKK β or the UC IKK β mutant was similar to endogenous IKK β levels (Figure 4D, lanes 4 and 7). While endogenous IKK β was proteolyzed significantly (Figure 4D, lanes 2 and 3), as was WT HA-IKK β (Figure 4D, lanes 5 and 6), the UC HA-IKK β mutant was resistant to proteolysis, and it even suppressed degradation of endogenous IKK β (Figure 4D, lanes 8 and 9). Of note, the sizes of wild-type HA-IKK β and the UC HA-IKK β mutant were reduced to that of endogenous IKK β due to the proteolysis of the HA-tag (also see Figures 3B and 4B). Similar results were also obtained in TNF- α -treated MCF-7 cells (Figure 4E).

Proteolysis of IKK β Eliminates Its Enzymatic Activity and Generates a Proapoptotic Factor

IKK β is essential for NF- κ B activation by TNF- α and plays a critical role in suppression of TNF- α -induced apoptosis (Li et al., 1999b; Tanaka et al., 1999; Li et al., 1999c). Physical destruction of IKK β by caspase-mediated proteolysis could result in detrimental effects on cell survival. To test this possibility, we constructed

a series of expression vectors encoding IKK β proteolytic fragments (Figure 5A). When expressed in HeLa cells, none of the fragments examined had any detectable kinase activity in response to TNF- α (Figure 5A), suggesting that proteolysis eliminated IKK β enzymatic activity. To determine the effect of IKK β fragments on cell survival, HeLa cells were cotransfected with expression vectors encoding HA-IKK β fragments and a transfection indicator green fluorescence protein (GFP) at a ratio of 4:1. Under these conditions, cells transfected with GFP also expressed HA-IKK β fragments (data not shown). Expression of HA-IKK β (1–546), but not other fragments, promoted TNF- α -induced apoptosis, as shown by Hoechst staining (Figure 5B).

One of the ways that HA-IKK β (1–546) promotes TNF- α killing is that it may interfere with IKK β activation. Indeed, expression of IKK β (1–546) partially inhibited activation of cotransfected M2-IKK β by TNF- α (Figure 5C). This inhibition was not the result of inhibiting expression of M2-IKK β , as shown by immunoblotting (Figure 5C). Under the same conditions, IKK β (1–373) had no inhibitory effect (Figure 5C), as did other fragments (data not shown). Consistently, expression of HA-IKK β (1–546), but not other fragments, partially blocked TNF- α -induced NF- κ B transcription activity, as measured by NF- κ B reporter gene assays (Figure 5D).

The Caspase-Resistant UC IKK β Mutant Suppresses Apoptosis Induced by TNF- α

If the proteolysis of IKK β by Casp3-related caspases is required for TNF- α -induced apoptosis, blocking IKK β proteolysis should suppress apoptosis. To test this scenario, HeLa cells were cotransfected with expression vectors encoding GFP and WT HA-IKK β , the UC HA-IKK β mutant, or empty vector. There was no difference between WT HA-IKK β and the UC HA-IKK β mutant with respect to their activation by TNF- α (Figure 6A). Immunoblotting analysis with anti-IKK β antibody showed that expression of the UC HA-IKK β mutant was similar to that of endogenous IKK β and HA-IKK β (Figure 6A). In parallel, cells were treated with TNF- α /CHX for 4 hr or were left untreated. The apoptotic cells, GFP-negative or -positive, showed typical apoptotic features, including membrane blebbing, cell shrinkage (Figure 6B, Phase panels), nuclear condensation, and DNA fragmentation (Figure 6B, Hoechst panels). Transfected cells were identified by fluorescence microscopy (Figure 6B, GFP panels). The death of GFP-positive cells differed significantly from each other (Figure 6B), while GFP-negative cells in each experimental group died at a similar rate (Figure 6B). WT HA-IKK β mildly suppressed TNF- α -induced apoptosis in HeLa cells (Figures 6B and 6C, HeLa panel, 16% less death than the control). In contrast, the UC HA-IKK β mutant significantly inhibited apoptosis (Figures 6B and 6C, HeLa panel, 52% less death than the control). Similarly, the UC HA-IKK β mutant suppressed TNF- α -induced apoptosis in MCF-7 cells (Figure 6C, MCF-7 panel, 40% less death than the control).

Since IKK β was also proteolyzed during Fas-mediated apoptosis (Figure 2D), we determined whether the UC HA-IKK β mutant was able to protect cells from Fas-mediated apoptosis. Indeed, expression of the UC IKK β

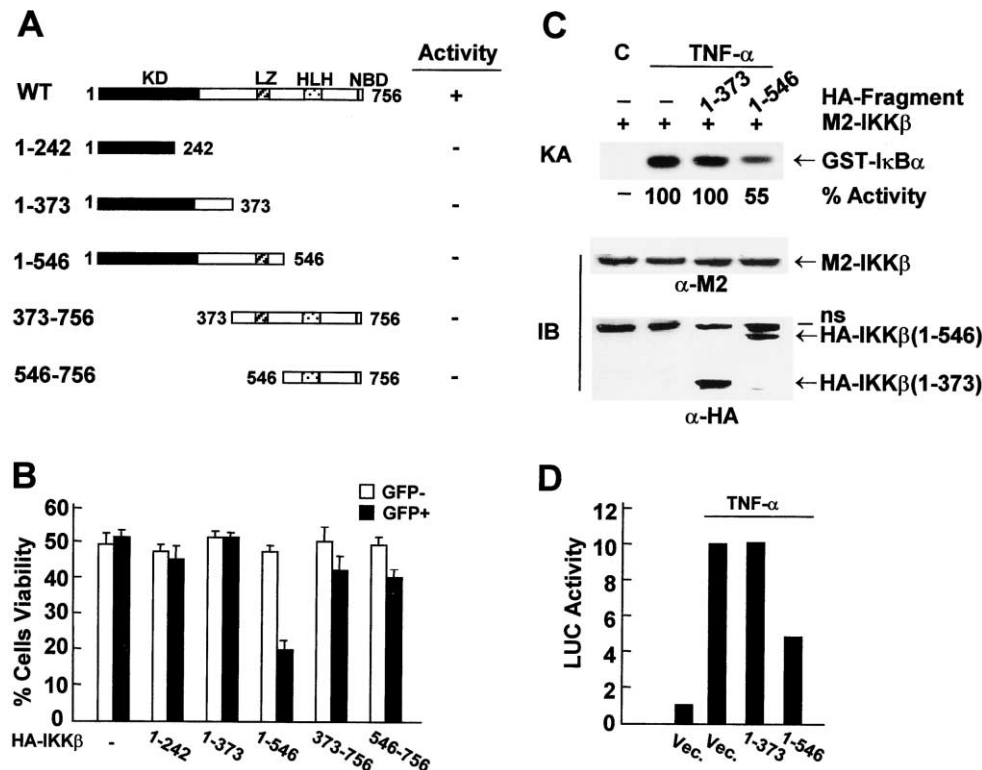


Figure 5. Proteolysis of IKK β Eliminates Its Enzymatic Activity and Generates an IKK Inhibitor that Promotes TNF- α -Induced Apoptosis

(A) Schematic presentation of truncated IKK β constructs and their enzymatic activity. (B) HeLa cells were transfected with various truncated HA-IKK β constructs or empty vector (4 μ g each), along with GFP (1 μ g). Cells were treated with TNF- α (10 ng/ml) and CHX (10 μ g/ml) for 4 hr and cell viability was analyzed by Hoechst staining. (C) HeLa cells transfected with M2-IKK β (1 μ g), along with HA-IKK β (1-373), HA-IKK β (1-546), or empty vector (4 μ g each) were treated with TNF- α (10 ng/ml) for 10 min. The activity of M2-IKK β and protein expressions were measured as described in Figure 1. (D) HeLa cells were cotransfected with a 2 $\times$ NF- κ B LUC reporter plasmid (0.5 μ g) and expression vectors encoding HA-IKK β (1-373), HA-IKK β (1-546), or empty vector (200 ng each). After 40 hr, cells were treated with TNF- α (40 ng/ml) for 8 hr or were left untreated. Relative LUC activity was determined as described (Lin et al., 1995). The results represent two separate experiments with similar results.

mutant significantly suppressed Fas-mediated apoptosis in HeLa and Fas-sensitive MCF-7 (MCF-7/Fas) cells (Figure 6D).

The UC IKK β Mutant Suppresses TNF- α -Induced Apoptosis by Maintaining a Significant Level of Antiapoptotic Proteins

One of the possible mechanisms by which the UC IKK β mutant suppresses TNF- α killing is that it can be activated over a prolonged period of time, leading to accumulation of antiapoptotic proteins. To test this possibility, HeLa cells were transfected with expression vectors encoding WT HA-IKK β , the UC HA-IKK β mutant, or empty vector. Cells were treated for various times with TNF- α in the presence of 10 μ g/ml CHX. Note that 10 μ g/ml CHX promoted TNF- α killing (Figure 6) but did not completely block synthesis of NF- κ B-induced proteins such as I κ B α (Figure 1F). Immunocomplex kinase assays revealed that IKK was initially activated at 15 min and reactivated again in all groups 3 hr later (Figure 7A). During this period of time, cells transfected with WT or the UC HA-IKK β mutant had comparable IKK activity (Figure 7A). The interval between activation and reactivation was about 2-3 hr, probably reflecting a refractory period caused by the internalization of TNF-R1 (Higuchi

and Aggarwal, 1994). Six hours after stimulation, IKK was reactivated only in cells transfected with the UC HA-IKK β mutant (6.3-fold) but not in cells transfected with empty vector or WT HA-IKK β (Figure 7A). Immunoblotting analysis with anti-IKK β antibody showed that expression of the UC IKK β mutant was similar to that of endogenous IKK β and WT HA-IKK β (Figure 7A). While endogenous IKK β and WT IKK β were significantly proteolyzed, the UC IKK β mutant was resistant (Figure 7A, also see Figure 4D). At this time point, the reactivated IKK had about 42% of its maximal activity in cells expressing the UC IKK β mutant (Figure 7A, panel UC HA-IKK β , compare lane 6 to lane 2). This was likely due to the loss of endogenous IKK β , which originally comprised approximately 50% of total IKK β proteins (Figure 7A, panel UC HA-IKK β , lane 2) and half of the total activity. Thus, expression of the UC IKK β mutant allows cells to have significant IKK activity for a prolonged period of time.

The persistent reactivation of IKK in cells expressing the UC HA-IKK β mutant did indeed lead to accumulation of antiapoptotic proteins controlled by NF- κ B. Immunoblotting analysis revealed that expression levels of c-IAP1 were still significant in cells transfected with the UC HA-IKK β mutant 6 hr after TNF- α /CHX treatment

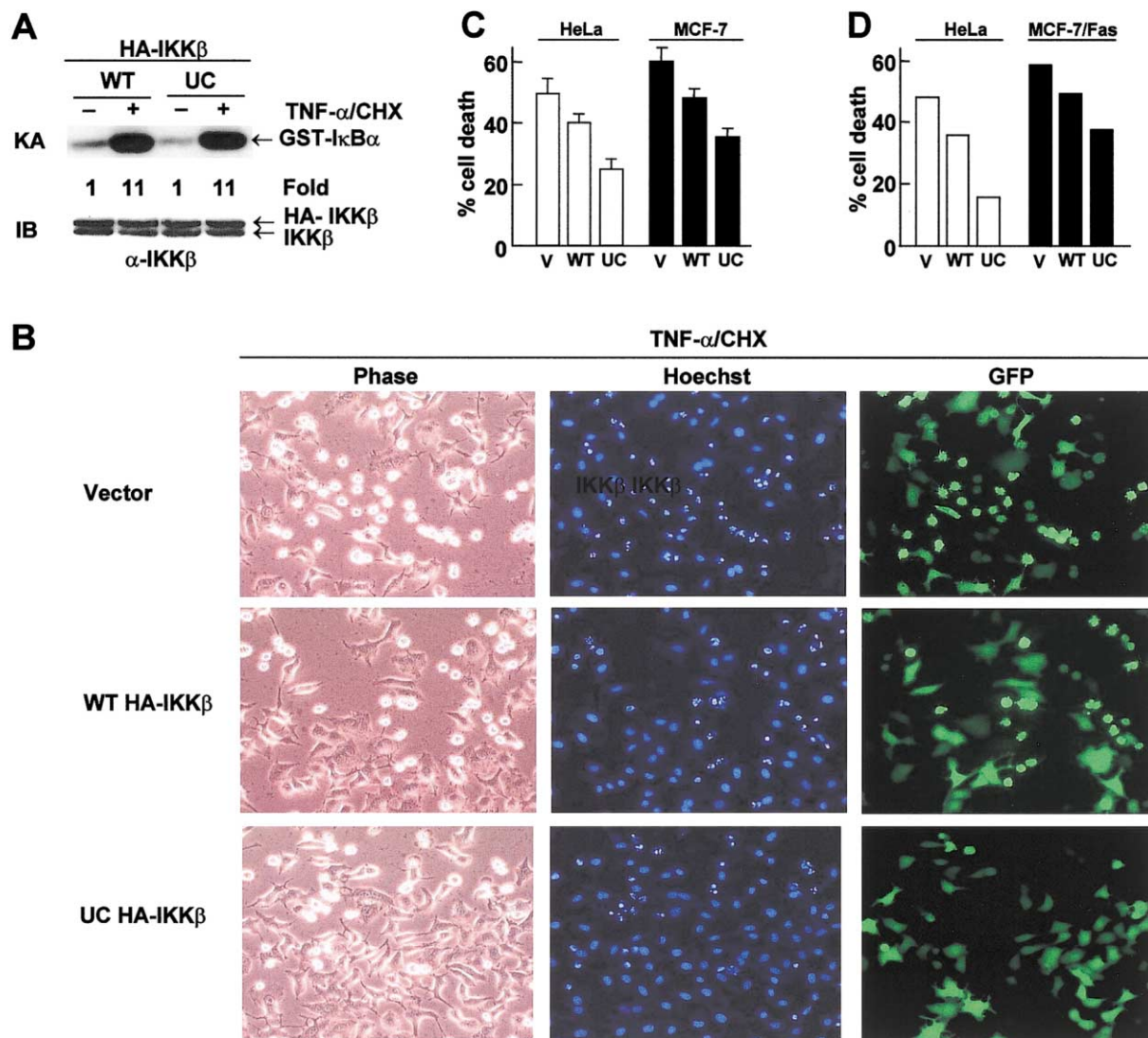


Figure 6. The UC IKK β Mutant Suppresses TNF- α Killing

HeLa, MCF-7, and MCF-7/Fas cells were transfected with expression vectors encoding WT HA-IKK β , the UC HA-IKK β mutant, or empty vector (4 μ g each) with a transfection marker GFP (1 μ g).

(A) HeLa cells were treated with TNF- α (10 ng/ml) and CHX (10 μ g/ml) for 15 min. The activity of WT or the UC HA-IKK β mutant was measured by immunocomplex kinase assays using anti-HA antibody. Expression levels of transfected and endogenous IKK β were analyzed by immunoblotting using the anti-IKK β C-terminal antibody.

(B) HeLa cells were treated with TNF- α (10 ng/ml) and CHX (10 μ g/ml) for 4 hr, stained with Hoechst, and visualized by light and immunofluorescence microscopy.

(C) HeLa or MCF-7 cells were treated with TNF- α and CHX (10 ng/ml, 10 μ g/ml) for 4 hr or TNF- α (10 ng/ml) for 11 hr, respectively. Cells were stained by Hoechst, and the death of the transfected cells (GFP positive) was calculated by counting eight random areas, and the differences in cell death were statistically significant ($p < 0.05$, Student's t test). The results are presented as mean $\pm$ standard error and represent three separate experiments.

(D) HeLa or MCF-7/Fas cells were treated with anti-Fas antibody (150 ng/ml) for 11 hr. The death of transfected cells was calculated as in (C). The data represents two separate experiments with similar results.

(Figure 7B, 72%). In contrast, c-IAP1 was diminished in cells transfected with vector or HA-IKK β (Figure 7B, 23% and 28%, respectively).

Discussion

Activation of NF- κ B suppresses apoptosis in many cell types (Liu et al., 1996; Beg and Baltimore, 1996; Van Antwerp et al., 1996; Wang et al., 1998). By mediating

NF- κ B activation, IKK plays a critical role in cell survival (Karin and Ben-Neriah, 2000). The current model for NF- κ B's antiapoptotic action is that activation of NF- κ B leads to expression of a group of antiapoptotic genes, such as c-IAP1, c-IAP2, TRAF1, TRAF2 (Chu et al., 1997; Wang et al., 1998), and XIAP (Wu et al., 1998), whose protein products are caspase inhibitors. This explains why TNF- α , which activates both NF- κ B and caspases, does not usually induce apoptosis unless NF- κ B activa-

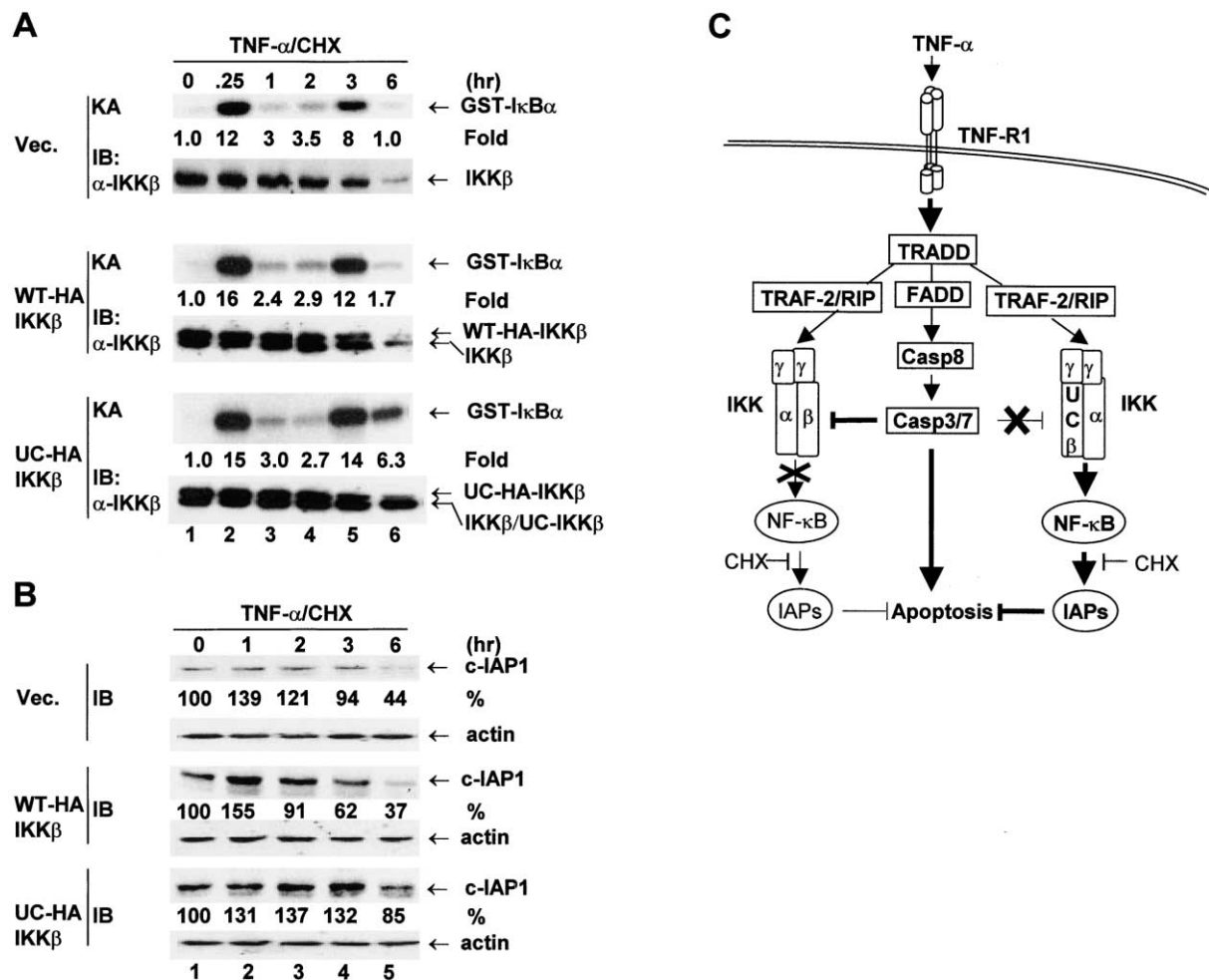


Figure 7. The UC HA-IKK β Mutant Inhibits TNF- α -Induced Apoptosis by Maintaining the Level of Antiapoptotic Proteins

HeLa cells were transfected with empty vector, wild-type HA-IKK β , or the UC HA-IKK β mutant (C) (5 μ g each). Cells were treated with TNF- α (10 ng/ml) and CHX (10 μ g/ml).

(A) IKK activity was measured by immunocomplex kinase assays. An aliquot of each sample was analyzed by immunoblotting using the anti-IKK β C-terminal antibody.

(B) The same samples in (A) were analyzed by immunoblotting with antibodies against c-IAP1 and actin. These experiments were repeated three times with similar results.

(C) Schematic illustration of the life and death balance in response to TNF- α . See the text for details.

tion is inhibited. Paradoxically, cells can undergo apoptosis in spite of the concomitant activation of NF- κ B by TNF- α (Chinnaiyan et al., 1995; Miura et al., 1995; Wang et al., 1998). It is unknown how cells override NF- κ B protection during apoptosis. In this report we demonstrate that proteolysis of IKK β by Casp3-related caspases eliminates NF- κ B protection, while a caspase-resistant IKK β mutant suppresses TNF- α killing. Thus, proteolysis of IKK β is a turning point in TNF- α -induced apoptosis.

The proteolysis of IKK β promotes TNF- α -induced apoptosis. Proteolysis of IKK β occurred in cells within 2 hr of TNF- α stimulation, almost at the same time as Casp8 and Casp3/7 activation (Figures 1 and 2). Thus, proteolysis of IKK β corresponded well with the onset of apoptosis and appeared to be an active, rather than a passive event. In vitro, IKK β was cleaved into multiple fragments by apoptotic cell extracts or purified Casp3

or Casp7 (Figures 3 and 4). Interestingly, only a portion of in vitro translated 35 S-labeled IKK β proteins was cleaved. It is possible that in vitro-translated IKK β proteins form aggregates that cannot be cleaved efficiently, a phenomenon that has been seen with some other in vitro-translated proteins (Lin et al., 1999; Takahashi et al., 1998). Two major fragments, p54(1–373) and p31(1–242) (Figure 4B), were detected in vivo by using the IKK β N-terminal antibody (Figure 2C). We were unable to detect other proteolytic fragments with either N-terminal or C-terminal IKK antibodies (data not shown), suggesting that those fragments are unstable intermediates in vivo. The proteolysis of IKK β in vivo is likely mediated by Casp3-related caspases (Figure 3E). We noted that anti-Casp7 antibody was also able to deplete the active fragment of Casp3 and vice versa (data not shown), which may explain why depletion of either Casp3 or Casp7 could block proteolysis of 35 S-labeled IKK β (Figures 3D

and 3E). This mutual depletion was specific since depletion with the anti-Casp8 antibody has no effect on IKK β cleavage (Figure 3E).

Proteolysis of IKK β abolished its kinase activity because none of its proteolytic fragments had detectable activity (Figure 5A). Furthermore, expression of IKK β (1–546) promoted TNF- α -induced apoptosis (Figure 5B), suggesting that proteolysis of IKK β not only eliminated a key survival factor but might produce a proapoptotic factor. The proapoptotic effect of IKK β (1–546) is likely due to its ability to inhibit TNF- α -induced activation of IKK (Figure 5C) and NF- κ B (Figure 5D). In support of this notion, other IKK β fragments with no detectable proapoptotic effects (Figure 5B) failed to block activation of IKK and NF- κ B (Figures 5C, 5D, and data not shown). We were unable to detect the endogenous 1–546 fragment, which is probably further proteolyzed in vivo (1–546 fragment can be further cleaved at Asp373, Asp242, and Asp78). It is conceivable, however, that the endogenous 1–546 fragment may promote TNF- α killing before it is further proteolyzed.

Is proteolysis of IKK β a decisive event in terminating NF- κ B protection during TNF- α -induced apoptosis? Recently, Liu and his colleagues have elegantly shown that RIP, which is involved in IKK activation (Devin et al., 2000; Zhang et al., 2000), is proteolyzed by Casp8 to promote TNF- α -induced apoptosis (Lin et al., 1999). It is not clear, however, whether proteolysis of RIP is the key event in terminating NF- κ B protection. In HeLa cells treated with TNF- α /CHX, activation of IKK and NF- κ B occurred much earlier than proteolysis of RIP (Figure 1 and data not shown) (Lin et al., 1999). Furthermore, IKK was reactivated again 3 hr later (Figure 7A), despite the partial proteolysis of RIP (data not shown). Proteolysis of IKK β directly eliminated IKK that had been activated and, more importantly, abrogated its reactivation at a later time point (Figures 7A, vector and WT IKK β panels, lane 6). Because activation of IKK is crucial for cell survival (Figures 6 and 7), proteolysis of IKK β is likely the decisive event in terminating NF- κ B protection. This notion is further supported by the fact that the caspase-resistant UC IKK β mutant (Figure 4) significantly suppressed TNF- α /CHX-induced apoptosis (Figure 6). In contrast, expression of a caspase-resistant RIP mutant induced apoptosis, although to a lesser extent than WT RIP (Lin et al., 1999). In addition to RIP and IKK β , several other components of the NF- κ B pathway, including TRAF1, TRAF2, NIK, and I κ B α , have been reported to be cleaved by caspases (Duckett and Thompson, 1997; Hu et al., 2000; Reuther and Baldwin, 1999). Whether proteolysis of these proteins is essential for terminating the NF- κ B protection in TNF- α killing has yet to be determined.

What is the mechanism by which the UC-IKK β mutant suppresses TNF- α -induced apoptosis? One of the possibilities is that it functions through prolonged induction of NF- κ B-controlled antiapoptotic proteins. We envision that apoptotic death of a cell requires amplification of the caspase cascade, an event that may depend on destruction of cellular survival factors. As illustrated in Figure 7C, treatment with TNF- α triggers rapid activation and reactivation of IKK and NF- κ B. This results in production of antiapoptotic proteins, such as c-IAP1, which inhibit caspases. The life balance is maintained and cells survive. The addition of CHX (10 μ g/ml) reduces the

production of antiapoptotic proteins, allowing activation of caspases such as Casp3. Activated Casp3 in turn cleaves IKK β , blunting IKK reactivation and subsequent production of antiapoptotic proteins. Activated Casp3 also cleaves antiapoptotic proteins, including c-IAP1 (Budihardjo, 1999). Therefore, the caspase cascade is amplified, switching the life balance toward death. In contrast, the UC IKK β mutant is resistant to caspase-mediated proteolysis and can be reactivated over a prolonged period of time (Figures 4D and 7A). Since 10 μ g/ml CHX reduces, but does not completely block protein synthesis, activation of NF- κ B by the UC IKK β mutant allows the accumulation of antiapoptotic proteins at a level that is sufficient to inhibit caspases. As a result, amplification of the caspase cascade and cell death is suppressed. We cannot, however, formally exclude the possibility that the UC IKK β mutant might suppress TNF- α killing through modulating additional substrates other than I κ B α that are involved in regulating apoptosis. Further experiments will be required to explore this possibility.

Is the proteolysis of IKK β critical for other types of apoptosis? Our results show that IKK β was cleaved during Fas-mediated apoptosis (Figure 2D), and expression of the UC IKK β mutant suppressed Fas-mediated cell death in both HeLa and MCF-7/Fas cells (Figure 6D). Since Fas does not activate IKK (data not shown), IKK appears to be a key player in cell survival, even in its unstimulated form. This is consistent with the observation that inhibition of NF- κ B sensitizes cells to Fas-induced apoptosis, although Fas engagement does not activate NF- κ B (Dudley et al., 1999). Thus, IKK β appears to play a critical role in the balance of life and death, and its proteolysis tips the balance toward death.

Experimental Procedures

Materials

Purified Casp3, Casp7, and Casp8 proteins, and antibodies against Casp3, Casp7, PARP, c-IAP1, and the C-terminal region of IKK β (629–735) were purchased from PharMingen. Anti-Casp8 antibody was kindly provided by Dr. Marcus E. Peter (Scaffidi et al., 1997). Rabbit polyclonal antibody against the first twelve amino acids of IKK β N terminus was produced by immunization of rabbits with LDH-conjugated synthetic peptide (Alpha Diagnostic International Inc., San Antonio). Antibodies against I κ B α , IKK α , IKK γ , Hemagglutinin (HA), and the C-terminal region of IKK β (470–756) were purchased from Santa Cruz. Anti-Fas antibody (CH11) was from Upstate. Cycloheximide, antibodies against M2, and actin were from Sigma. hTNF- α was purchased from R & D Systems. z-VAD-FMK, DEVD-FMK, and YVAD-CMK were from CalBiochem. In vitro transcription and translation kit (TNT) was from Promega. [γ - 32 P]ATP (3000 mCi/nmol) and [35 S]methionine were from Dupont NEN.

Cell Culture, Transfection, and Transcription Assays

MCF-7 (a gift of Dr. Zheng-gang Liu) and MCF-7/Fas (a gift of Dr. Marcus E. Peter) cells were grown in RPMI 1640 medium, while HeLa cells were cultured in Dulbecco's modified Eagle medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 2 mM glutamine, 100 U/ml of penicillin, and 100 μ g/ml of streptomycin. Transfections were performed using the Ca $^{2+}$ /phosphate method and 2 $\times$ NF- κ B-luciferase reporter gene activity was determined as described (Lin et al., 1995; Nemoto et al., 1998).

Plasmids

Mammalian expression vectors of HA-IKK β and M2-IKK β have been described previously (Nemoto et al., 1998). The HA-tagged putative caspase-resistant IKK β mutants were constructed by site-directed mutagenesis using the Quick-exchange mutagenesis kit (Stra-

tagene). The HA-tagged proteolytic IKK β fragments were generated by PCR and subcloned into pcDNA3.1 vector (Invitrogen). All constructs were confirmed by sequencing.

In Vitro Proteolysis Assays

HA-IKK β and its putative caspase-resistant mutants were translated in vitro in the presence of [³⁵S]methionine using the TNT system (Promega). For preparation of apoptotic HeLa cell extracts, cells were treated with TNF- α (20 ng/ml) and CHX (10 μ g/ml) for 4 hr and then resuspended in hypotonic buffer (10 mM Hepes [pH 7.0], 40 mM glycerophosphate, 50 mM NaCl, 2 mM MgCl₂, 5 mM EGTA, 1 mM DTT, 1 mM PMSF, and 0.5 μ g/ml aprotinin). After freezing and thawing, the lysates were cleared by centrifugation and the supernatants were aliquoted and stored at -80°C. To deplete caspases from apoptotic extracts, the extracts (50 μ g) were incubated with their corresponding antibody and Protein A-Sepharose beads for 4 hr at 4°C. The caspase-depleted supernatants were used for in vitro proteolytic assays. The proteolysis of the ³⁵S-labeled HA-IKK β proteins (1 μ l) by purified recombinant caspases (200 ng), control, or apoptotic extracts (50 μ g) was carried out at 37°C for 2 hr in the presence or absence of various caspase inhibitors as indicated in the Figure legends. To inhibit caspases in vitro, purified caspases or apoptotic extracts were incubated with specific caspase inhibitors for 15 min prior to incubation with ³⁵S-labeled IKK β proteins in the proteolysis assay.

Protein Kinase Assays and Immunoblotting

Endogenous or transfected IKK β was immunoprecipitated from MCF-7 or HeLa cell extracts with the corresponding antibody, as indicated in Figure legends. The kinase activity of the immunocomplex was performed as described (Nemoto et al., 1998). Radioactivity in the phosphorylated proteins was quantitated using a Phosphorimager. Immunoblotting analysis was performed as described (Nemoto et al., 1998). The antibody-antigen complexes were visualized by the enhanced chemiluminescence (ECL) detection system (Amersham) and quantitated using a Chemiluminescence 5500 (Alpha Innotech).

Apoptosis Assays

HeLa, MCF-7, or MCF-7/Fas cells were cotransfected with various IKK β constructs along with green fluorescent protein (GFP) at a ratio of 4:1. Cells were treated with TNF- α (10 ng/ml) in the presence or absence of CHX (10 μ g/ml), or anti-Fas antibody (150 ng/ml) 40 hr posttransfection. At various times, cells were stained with either Hoechst or trypan blue, and the death of nontransfected and transfected cells was analyzed by fluorescence or light microscopy, respectively.

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